

Education

University of Toronto

September 2024 - December 2025 (Expected)

MSc in Applied Computing, Applied Mathematics stream.
2024 Vector Scholar in Artificial Intelligence (\$17,500)
Coursework: Generative NN, Nonlinear Numerical Analysis, Physics-Based Animation, Physics-Informed Neural Representations for Visual Computing, Communication for Computer Scientists, Technical Entrepreneurship.

University of Toronto

September 2020 - June 2024

HBSc Mathematics and Physics Specialist. High distinction.
Coursework: Machine learning, Deep learning, numerical analysis, optimization, real analysis, complex analysis, PDEs, linear algebra, Fourier analysis, probability, topology, differential geometry, nonlinear physics, quantum information theory.

Experience

Deep Learning Researcher · Vector Institute

November 2023 - Present

- Discovered novel phenomenon coined Transformer Alignment (TA), displayed positive correlation with generalization.
- Performed extensive experimentation on 38+ open-source Huggingface LLMs.
- Submitted work to ICML and NeurIPS 2024. See a [preprint](#) here.
- Supervised by Professor Vardan Papyan.

Algorithms Researcher and Designer · University of Toronto

April 2023 - Present

- Developing algorithms for solving scalar differential equations and for numerical integration.
- Our methods dominate over alternative methods in the high-frequency regime.
- Independently decreased time complexity by a factor of n by exploiting an over-determined system.
- Proved algorithm correctness by establishing theoretical bounds on computation error.
- Establish theoretical framework, implement, test and iteratively optimize, publish results in academic journals.
- Supervised by Professors James Bremer and Kirill Serkh.

Teaching Assistant · University of Toronto

September 2023 - Present

- Run tutorials, host office hours, mark and invigilating exams.
- Fall 2024: [MAT292](#), ODEs for second-year engineering students.
- Winter 2024: [MAT187](#), Calculus 2 for first-year engineering students.
- Fall 2023: [MAT244](#), ODEs for second-year math major students.

Machine Learning Researcher/Engineer · CERN × University of Toronto

September 2023 - April 2024

- Trained a graph neural network to detect tau neutrinos and their decay products.
- Combatted an imbalanced dataset by implementing a modified, weighted loss function.
- Improved classification performance on underrepresented training samples by over 20%
- Supervised by Professor Nikolina Ilic, collaborated with her CERN research team of graduating students.

Tutor · Forest Hill Tutoring

April 2023 - August 2024

- High school math and physics tutoring, largely focused on helping grade 11 and 12 IB students.
- Principal instructor for Ontario credited courses such as MDM4U, MHF4U and MCV4U.

Structural Engineering Assistant · Safeline Construction Group

April 2022 - October 2022

- Worked on a 57 storey building in downtown Toronto assisting the senior and assistant supervisors.
- Independently managed interior renovation of a small townhouse in downtown Toronto.
- Worked alongside mechanical, civil, and electrical engineers throughout entire construction process.

Awards, Publications & Talks

- Vector Scholarship in Artificial Intelligence.** 2024
Awarded by the [Vector Institute](#).
A \$17,500 entrance scholarship awarded to top students pursuing AI master's in Ontario universities.
- A Solver for Linear Scalar ODEs Whose Running Time is Bounded Independent of Frequency.** *Preprint.* 2023
Murdock Aubry, James Bremer. [PDF](#).
- Transformer Alignment in Large Language Models.** *Preprint.* 2024
Murdock Aubry, Hoaming Meng, Anton Sugolov, Vardan Papyan. [PDF](#).
- On the Two Dimensional Levin Method.** *Preprint.* 2024
Murdock Aubry, Kirill Serkh, James Bremer. [PDF](#).
- The Emergence of Clusters in Self-attention Dynamics.** *Talk.* 2023
UofT Graduate Applied² Math Seminar. [Slides](#).
- Development of Graph Neural Networks for Tau (τ) Neutrino Detection.** *Talk.* 2023
CERN and Deep Underground Neutrino Experiment (DUNE). [Slides](#).
- On the Adaptive Levin Method.** *Talk.* 2023
Canadian Undergraduate Mathematics Conference. [Slides](#).

Technical Skills

- Languages.** Python, MATLAB, Fortran, HTML, CSS, SQL, Bootstrap, Bash, Mathematica, $\text{\LaTeX}$, Markdown.
- Packages.** PyTorch, Transformers, TensorFlow, Numpy, Scipy, CUDA, Pandas, Matplotlib, Seaborn, JSON, OS, Selenium, BeautifulSoup, ezdxf, dxfgrabber.
- Software.** Git, SSH, SLURM, Tensorboard, AutoCAD, Unix.